Landscape-scale bacterial biogeography across the Rub’ al-Khali reveals recurring spatial and soil-position structure

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ABSTRACT

The Rub’ al-Khali is the world’s largest continuous sand desert, yet the bacterial ecology of its open Saudi interior has not been surveyed. We present the first landscape-scale bacterial survey of this region, with 5 expeditions to 60 sites over 31 months and paired surface, shallow-subsurface and root-adjacent soil. Bacterial communities formed a recurring geographic pattern: distant sites contained more different communities, mainly because taxa replaced one another across the landscape. Within sites, root-adjacent communities were distinct and less even than both bulk-soil positions, showing that local soil context changes which bacteria dominate. Long-term temperature, rainfall and humidity tracked diversity and the relative abundance of many genera, while richness showed a short positive association with recent rain, strongest during the first several days. Soil pH accounted for part of the geographic pattern. The metabolic pathways predicted from the 16S sequences retained both geographic and soil-position structure, and their broad gene-family profiles agreed with matched shotgun data. Checks for DNA from damaged cells and for low-biomass contamination showed that these possible assay effects did not change the main ecological conclusions. The study establishes a regional biological baseline and a reusable resource for biodiversity monitoring, restoration, arid-agriculture research and evidence-based land planning.

Keywords: Rub’ al-Khali, Empty Quarter, hyperarid soil microbiome, bacterial biogeography, landscape-scale survey, soil position, repeated sampling, 16S rRNA amplicon survey

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INTRODUCTION

The Rub' al-Khali (Empty Quarter) is the world's largest continuous sand desert. It covers much of the southern Arabian Peninsula, mainly within Saudi Arabia [1], and shapes land use, conservation and plans for agriculture across the region. Yet the bacteria in its open interior have not been surveyed. The closest molecular community study examined a groundwater-fed salt flat in the United Arab Emirates, a habitat unlike the open sand soils of the Saudi interior [2]. A regional survey is needed both to understand this major desert and to provide a reference for future monitoring, restoration and arid-land research.

Work in other deserts shows why such a survey matters. Dry landscapes that look uniform can contain strong changes in bacterial composition. Climate, soil pH, salts and minerals are often associated with these changes [3, 4, 5, 6]. Rare rain can also release a short biological pulse before the soil dries again [7, 8]. Sampling sites at different delays after rain can therefore test whether diversity rises and then declines, although it cannot identify the process. One visible spatial outcome is distance decay: nearby sites contain more similar communities than distant sites [9, 10, 11]. This pattern can arise because taxa replace one another across the landscape, or because poorer communities become subsets of richer ones. Replacement implies distinct local assemblages; nestedness implies loss from a shared pool. Neither process has been measured across the open Empty Quarter.

Desert soil can also change within centimetres. The surface receives direct sunlight and strong heating, while soil at 5–10 cm is partly buffered. Sparse plants may add water and carbon near their roots [12, 13, 14]. These differences can change which taxa occur, which taxa dominate, and which metabolic pathways the community is predicted to encode. Richness alone cannot separate these outcomes: two soils may contain similar numbers of taxa while their abundance and predicted functional profiles differ. DNA can also remain after cells die. This relic DNA, together with contamination introduced during laboratory processing, is a particular concern in low-biomass desert soil. Biological interpretation therefore requires controls that distinguish these sources of DNA from the sampled community [15].

Arabian studies have described bacteria along elevation gradients, at single arid locations, and in plant-associated soil [16, 17, 18, 13, 19, 14, 20]. None covers the open Rub' al-Khali at landscape scale or combines a dense transect with repeated sampling of several soil positions. We therefore sampled 60 sites during 5 expeditions over 31 months. We asked how bacterial composition changes across the landscape and among soil positions; how climate, pH and elemental composition relate to those patterns; whether predicted metabolic potential changes with them; and whether relic-DNA and low-biomass controls alter the main conclusions. The resulting survey maps bacterial biogeography, meaning the distribution of communities, at landscape and centimetre scales. A linked data resource preserves the samples, measurements and sequences, together with records of how they were produced, for reuse [21].

RESULTS

Bacterial communities change across the landscape

We surveyed 60 core sites along a west–east transect through the open Saudi Rub' al-Khali and returned during 5 expeditions over 31 months (Figure 1a). At each visit, we sampled surface, shallow-subsurface and root-adjacent soil. Sequencing a marker region of the bacterial 16S rRNA gene yielded 1,237 quality-controlled profiles; 1,227 came from the repeatedly sampled core sites (Figure 1b). Trip 2 reached only 8 sites because of extreme heat. Supplementary Table S1 gives the samples used in each analysis.

We first asked whether community composition changed gradually across the route. The model allowed two broad patterns: a steady change from west to east and a smooth change in its rate that could reverse direction once. These were represented by a straight-line term and a squared-location term. Together, these patterns accounted for 40.2 % of the site-averaged difference in relative taxon abundance across 630 site, expedition and soil-position combinations (95 % CI, 32.5–47.8 %; $p = 0.001$). The result remained when we changed the number of genera, used exact marker sequences, called amplicon sequence variants (ASVs), as the taxonomic units, or omitted any one expedition (Supplementary Section S5.3). The geographic pattern therefore did not depend on grouping sequences into genera or on a single visit.

Communities also became less similar with distance. We therefore asked a more direct question: did sites farther apart have more different relative taxon abundances? We measured this with Aitchison dissimilarity, which compares the ratios among taxa rather than raw read totals. It is zero when two relative compositions are identical and increases as their taxon ratios diverge [22]. A positive slope with distance therefore shows how rapidly community composition changes across the landscape. The slope was 1.351 (95 % CI, 0.905–1.796), 1.510 (1.092–1.929) and 1.180 (0.698–1.662) dissimilarity units per 100 km in surface, shallow-subsurface and root-adjacent soil, respectively (each $p = 0.0001$; Figure 1d). The slopes differed among positions ($p = 0.0053$): composition changed fastest below the surface and slowest next to roots.

Finally, we asked what kind of change produced these differences. After comparing every sample at the same read coverage, we used Sørensen dissimilarity to compare the proportions of shared and unshared taxa between two sites. It is zero for identical taxon lists and increases as fewer taxa are shared. We separated taxa that replaced one another between sites from taxa lost from an otherwise shared pool; these components are called replacement and nestedness, respectively [23]. Replacement accounted for 69–75 % of the average difference in every soil position. Both components rose with distance, but replacement dominated. The Empty Quarter is therefore a mosaic of distinct bacterial assemblages rather than a simple decline from one shared community.

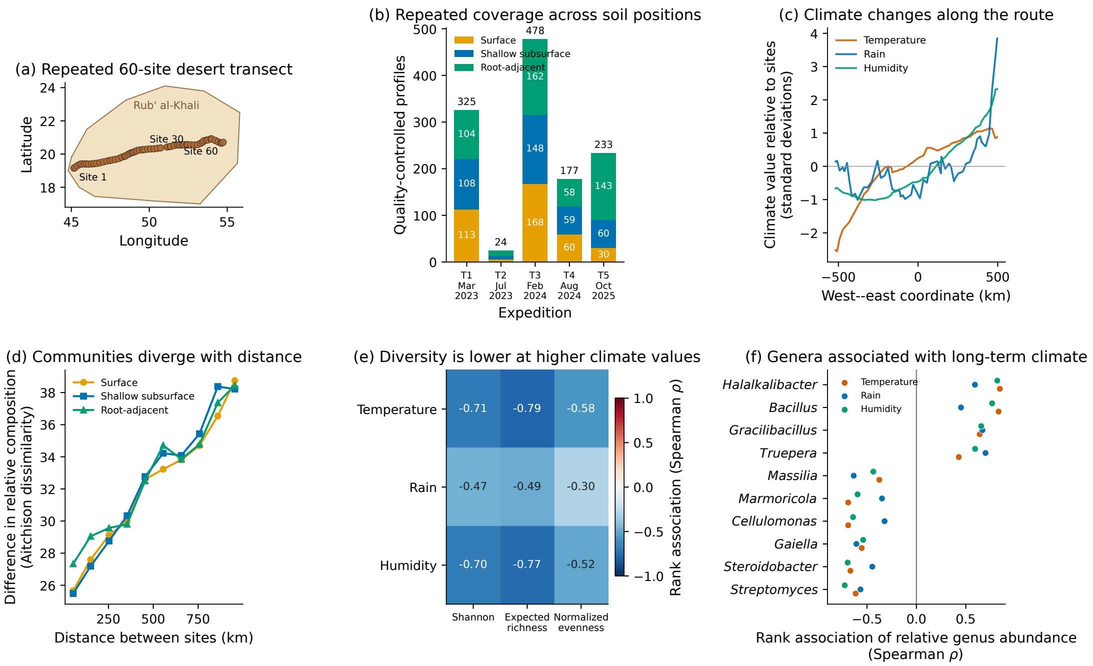

Figure 1. Survey design, geographic differences and climate associations. (a) The 60 core sites along the west–east route. The shaded Rub' al-Khali boundary is for orientation only. (b) The 1,237 quality-controlled profiles by expedition and soil position. Trip 2 reached 8 sites. (c) Each site's 49-month climate mean, expressed as standard deviations from the mean across all sites. (d) The difference in relative community composition, measured as mean Aitchison dissimilarity, as distance between sites increases. The statistical test kept all comparisons involving the same site together. (e) Rank associations between climate and diversity. Values near -1 mean that diversity consistently decreases as the climate variable increases; values near zero mean no consistent order. (f) The strongest supported genus associations. Tests in (e,f) were corrected for the number of comparisons.

Soil position shapes bacterial composition and evenness

We next asked whether soil position changed the relative abundance of taxa within the same site and expedition. Across 170 site-by-expedition comparisons in which all three soil positions were available, the separation among the three average compositions was large relative to the variation within positions. This ratio is the pseudo-F statistic: a larger value means stronger overall separation. The soil-position separation had pseudo-$F = 17.28$ ($p = 0.001$). We then asked whether sites shifted in a consistent direction for each position pair. We divided the magnitude of the average shift by the average magnitude of the individual site shifts. This standardized displacement is near 1 when sites shift in the same direction and near 0 when their shifts cancel. Root-adjacent and shallow-subsurface soil showed the most consistent shift (0.478), followed by root-adjacent and surface soil (0.433) and shallow-subsurface and surface soil (0.331; Figure 2a). All 3 comparisons passed correction for the 3 tests and remained supported after changing the included taxa, the treatment of zero counts, or the omitted expedition (Supplementary Section S5.1).

Shannon diversity combines the number of taxa with the balance of their read abundances. It was lower in root-adjacent than shallow-subsurface soil by 0.206 on average (Figure 2b). After resampling whole sites, the 95 % interval was [−0.398, −0.001]. The difference remained after adjustment for sequencing depth and recorded extraction method (−0.273, 95 % CI [−0.424, −0.122]).

To find what produced this change, we estimated how many taxa would be expected if every profile had 25,000 reads. We then scaled Shannon diversity by this common-depth richness to separate the balance among taxa from their number. This measure is normalized evenness: a high value means that reads are spread across taxa, whereas a low value means that a few taxa dominate. Evenness declined from shallow-subsurface to surface and then to root-adjacent soil (Figure 2c). Root-adjacent soil was lower than surface soil by 0.0315 (95 % CI [−0.0448, −0.0186]) and lower than shallow-subsurface soil by 0.0440 ([−0.0572, −0.0313]). Expected richness did not show the same order. Soil position therefore changed which taxa dominated more consistently than it changed how many taxa were detected. Because the read abundances are relative, we compared each genus with the geometric mean abundance across all genera on a logarithmic scale. This avoids treating the reads as absolute cell counts. The largest descriptive genus shifts involved contrasts such as *Pelagibacterium*, *Bradyrhizobium*, *Deinococcus* and *Planococcus* (Figure 2d). These values locate the genera that contributed most to the overall composition difference; they do not test each genus separately.

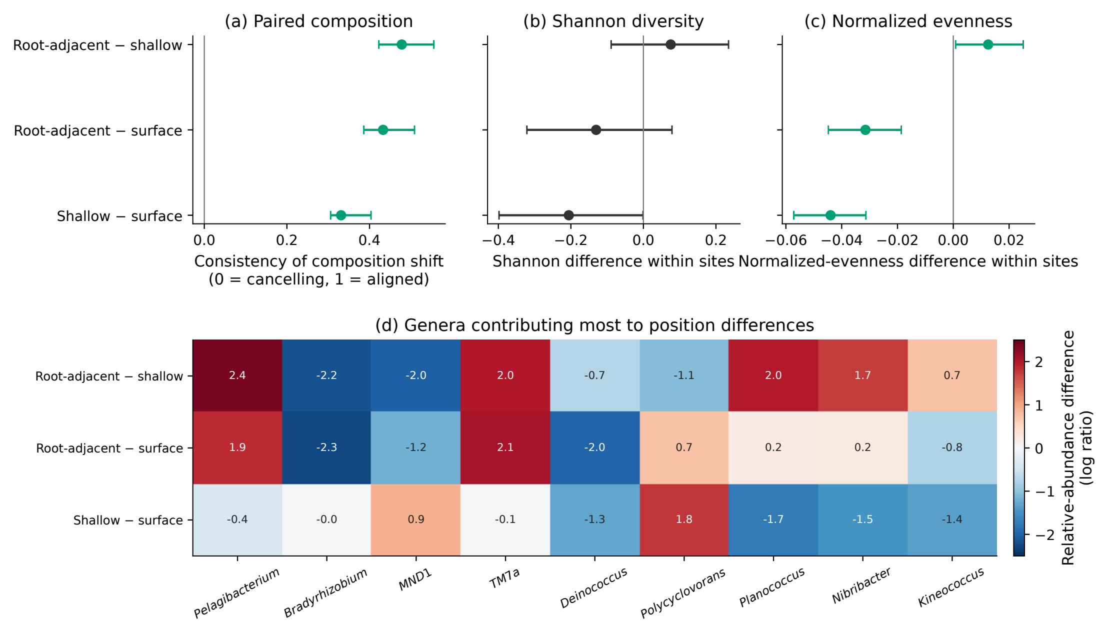

Figure 2. Soil position differentiates bacterial communities within sites. (a) The consistency of the shift in relative taxon abundance across sites. Values approach one when sites shift in the same direction and zero when shifts cancel. (b,c) Paired differences in Shannon diversity and normalized evenness, averaged across expeditions. In (a–c), points show means and bars show 95 % intervals. (d) Genera that contributed most to the 3 overall composition differences. Positive values mean a higher relative abundance in the first named position. This panel describes contributions to the overall difference and does not test genera separately.

Climate and soil properties track bacterial variation

We recorded air temperature, pressure and relative humidity during 247, 177 and 239 quality-controlled site visits. Air temperature ranged from 20.0 to 49.4 °C and humidity from 9.5 to 62.5 %. To describe background climate rather than conditions at one visit, we also summarized 49 months per site from January 2022 to January 2026. Long-term site means ranged from 26.38 to 30.30 °C, 1.31 to 7.47 mm rain per month, and 24.27 to 35.40 % relative humidity. Figure 1c shows how these site means changed across the transect.

We asked whether sites ordered from lower to higher climate values also showed a consistent order in diversity. Spearman rank correlation measures this ordering: a value near -1 means that diversity consistently decreases as the climate value increases, whereas a value near zero means no consistent order. All 3 long-term gradients tracked diversity (Figure 1c–f). We compared 3 aspects of diversity: Shannon diversity combines taxon number and abundance balance, expected richness estimates how many taxa would be detected at a common read coverage, and normalized evenness measures how evenly reads are spread after accounting for richness. Sites with higher temperature, rainfall and humidity had lower Shannon diversity (Spearman $\rho = -0.71$, -0.47 and -0.70), lower expected richness (-0.79, -0.49 and -0.77), and lower evenness (-0.58, -0.30 and -0.52). All 9 associations remained supported after correction for the 9 comparisons (adjusted $p \leq 0.021$). Among 200 genera detected in at least 20 % of profiles, 112, 112 and 111 were associated with temperature, rainfall and humidity after multiple-testing correction. Supplementary Section S7 gives the genus rankings and shows how strongly these climate patterns overlap with location.

Quality-controlled pH measurements matched 702 profiles and had a mean pH of 7.986. After accounting for expedition and soil position, location along the route explained 84.6 % of pH variation ($p \leq 0.001$). Among the same 702 matched profiles, the share of bacterial composition explained by location fell from 35.8 % to 19.6 % after pH adjustment. Excluding the group with the highest pH, or its entire site, gave 18.7–19.6 %. pH and location therefore described substantial overlapping variation.

X-ray fluorescence (XRF) measured 11 elements in at least 75 % of 725 specimens. Because the elements changed together, we combined them into one summary gradient that captured the largest shared change. This gradient represented 39.2 % of the elemental variation and a smaller but consistent share of bacterial composition, 0.358 % ($p = 0.010$). Planned changes to the included taxa and treatment of zero values gave estimates from 0.311 to 0.543 % (Supplementary Sections S2–S3). This range shows sensitivity to analysis choices; it is not a confidence interval.

Bacterial richness shows a short association with recent rain

We next asked whether rare rain was followed by a temporary change in diversity. We combined rain during the preceding 60 complete days into a pulse score. The score represents an association that rises after rain, reaches an unknown peak, and then fades. We tested possible peaks from 0.5 to 30 days for 3 diversity measures while accounting for site, soil position, expedition and location along the route. To test whether the strongest observed association could arise from searching across measures and delays, we repeatedly shifted the observed rain timing within each expedition while preserving its storms. This randomization covered the complete search and does not require storms to recur across expeditions.

Expected richness was selected with both daily rainfall datasets. In one independent product [24], each additional millimetre at the fitted peak was associated with 327.5 additional expected taxa (95 % site-clustered CI, 141.1–514.0). The fitted peak was 1 complete day (95 % peak-selection interval, 0.5–6.5 days), and rain accounted for 3.84 % of the remaining variation (0.57–8.99 %). The family-wise one-sided test for a positive pulse gave $p = 0.0432$. In the other product [25], the corresponding estimate was 179.6 expected taxa per millimetre (95 % site-clustered CI, 92.1–267.1), with a peak at 2 days (1.5–4.5 days), 3.64 % explained variation (0.60–7.87 %), and $p = 0.0560$ after the same correction.

The pulse curve assumes that an association rises and then declines smoothly. As a check on this shape, we fitted rain from 7 non-overlapping periods before collection together for each dataset. Both fits placed their largest positive estimate at 3–4 days. The fitted association was positive in every soil position and changed little after accounting for pH or candidate contaminants, but weakened when the model allowed more complex geographic change. The data therefore support an early association between rain and richness, not an exact delay or a causal mechanism (Supplementary Section S7).

Predicted metabolic pathways follow geography and soil position

We used PICRUSt2, which compares each 16S sequence with reference genomes, to predict which metabolic pathways the community could encode [26]. These estimates describe metabolic potential, not activity. Across 1,227 core-site profiles and 462 pathways in the MetaCyc database [27], the 200 most abundant pathways retained a strong geographic pattern. The same route model, which allowed the rate of change to vary smoothly and reverse once, accounted for 23.4 % of the difference among site-level pathway profiles (95 % CI, 15.5–31.4 %; $p = 0.0001$). The result remained supported after changing the included pathways, the treatment of zero values, or the omitted expedition.

We then asked whether predicted pathway profiles differed among soil positions, and found a difference across 172 complete site-by-expedition comparisons ($p = 0.0001$). We used the same standardized displacement as for taxonomic composition. Predicted pathways shifted most consistently between root-adjacent and shallow-subsurface soil (0.453) and between root-adjacent and surface soil (0.429; both adjusted $p = 0.00015$ after correction for 3 comparisons; Figure 3a). Of 600 tests covering 200 pathways and 3 position pairs, 270 remained supported after correction. They included pathways for amino acids, cofactors, carbohydrates and cell envelopes; Supplementary Section S8.1 gives the complete results.

From west to east along the transect, 2 pathways that produce the fatty acids cis-vaccenate and gondoate increased, while palmitate production decreased. Root-adjacent soil had higher predictions for glucose and xylose degradation and for several polyamine and thiamin pathways. These patterns identify candidates for tests of membrane adaptation and carbon and metabolite exchange near desert roots; they do not show that these processes occurred.

Prediction quality depends on how closely the sampled bacteria resemble organisms with sequenced reference genomes. PICRUSt2 summarizes this distance as the weighted nearest-sequenced-taxon index (NSTI); smaller values require less extrapolation, and the median here was 0.165. We also made a more direct comparison in 125 samples. For each sample, we ranked 3,471 gene families by abundance in the PICRUSt2 prediction and in shotgun metagenomic data. Shotgun sequencing reads all recovered DNA rather than one marker region, so it measures gene families more directly than a 16S-based prediction. These families, called KEGG Ortholog groups, collect genes with the same broad function. A rank correlation near one means that the two methods place the gene families in nearly the same order. The median correlation was 0.743 (95 % interval, 0.731–0.751, after resampling whole sites), and the correlation between the two community-average profiles was 0.806 (0.793–0.812; Figure 3b). This level of agreement supports broad functional comparisons while leaving individual pathways as predictions [26, 28]; Supplementary Section S8.2 gives the matched-data analysis.

DISCUSSION

The Empty Quarter and global dryland ecology

The open soils of the Saudi Rub' al-Khali are organized both across hundreds of kilometres and within a few centimetres. Taxa replace one another across the landscape, while soil position changes community composition and the balance among taxa within sites. Climate, pH and elemental composition place this variation in an environmental setting, and predicted pathways suggest that the taxonomic changes carry differences in metabolic potential. The main lesson is not that one variable controls the desert, but that a hyperarid sand sea contains a repeatable mosaic at landscape and local scales.

Geographic community change recurs across desert microbiomes. In the Empty Quarter, the transect pattern remained at genus and ASV resolution and across expeditions. Taxon replacement formed 69–75 % of mean dissimilarity, so distant sites usually held different assemblages rather than smaller subsets of one pool. Distance decay also occurs in Negev soils and Namib dune systems [11, 29]. Different primers, distances and soils prevent numerical comparison, but all three deserts reject the hypothesis that there is one representative "desert soil" community.

Water was associated with diversity at two scales. Along the landscape, long-term temperature, rainfall and humidity were negatively associated with diversity. After individual rain events, by contrast, richness was highest during the first several days and the fitted association then declined. These patterns answer different questions: the long-term comparison separates sites with different climate histories, while the pulse model compares rainfall amount and time before collection. Pulse-reserve theory predicts that rare rain can briefly release water limitation before drying restores it [7, 8]. Rapid cell resuscitation, movement of solutes, or improved detection of rare taxa could each produce our pattern; the marker-gene data alone cannot distinguish them. Direction also varies among deserts. In a public Atacama gradient, soil humidity was positively associated with Shannon diversity even after accounting for latitude, longitude and elevation (rank correlation 0.68, $p = 0.004$) [30]. Water source, evaporation, soil chemistry and the range sampled therefore matter more than a global rule such as "more water means more diversity" [31].

Physical stratification also recurs, although its direction depends on scale. An Atacama pit showed differences in community composition among depth zones ($p = 0.0014$) without a steady increase or decrease in diversity, and Mojave biocrust communities differ between surface and subsurface soil [12, 32].

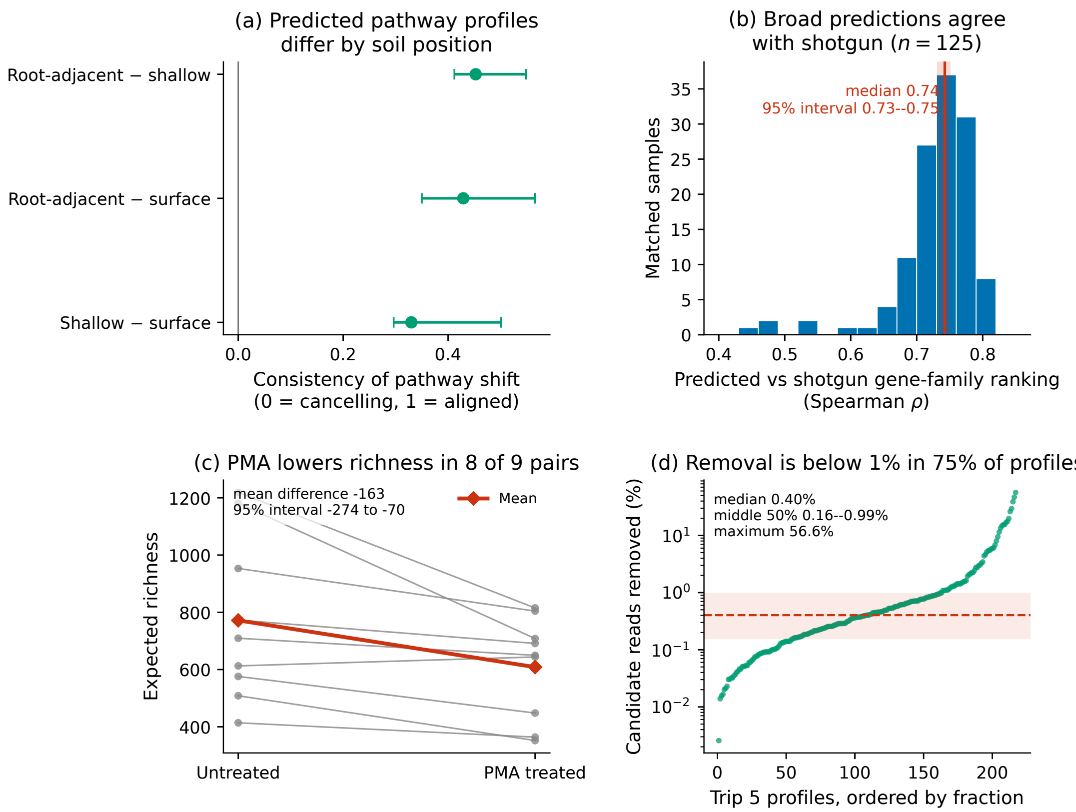

Figure 3. Predicted function and assay controls. (a) Paired differences in predicted pathway profiles. Values approach one when sites shift in the same direction and zero when shifts cancel; bars show 95 % intervals from resampling whole sites. (b) Agreement between predicted and shotgun-measured gene-family rankings; red marks the median and its 95 % interval. (c) Estimated richness before and after PMA treatment; red joins the means and the annotation gives the interval for their difference. (d) The fraction of reads assigned to candidate contaminants; red marks the median and the range containing the middle half of profiles.

The Empty Quarter adds repeated comparisons of surface, 5–10 cm and root-adjacent soil. Root-adjacent communities were less even rather than simply poorer, so fewer taxa carried more reads. This pattern is consistent with a stronger selective environment near roots: plants change carbon and nutrient supply, and studies of desert plants have found host- and root-compartment-specific recruitment [33, 13, 34, 20]. However, our loose root-adjacent soil and missing host identities do not show that the plant caused the difference. The genus contrasts in Figure 2d identify targets for culture, genome and plant-association studies.

Predicted pathways followed geography and soil position, suggesting that the taxonomic mosaic carries different metabolic potential. Agreement with matched shotgun profiles supports this broad reading, not measured activity. In the Namib, direct gas uptake, ATP and respiration measurements showed more than a 30-fold rise in hydrogen oxidation after wetting [35]. Our predictions identify where comparable process measurements are needed.

The relic-DNA experiment addresses DNA that remained accessible after cell membranes were damaged. PMA treatment lowered expected richness in 8 of 9 pairs (Figure 3c), showing that PMA-sensitive DNA contributed taxa to these aliquots. PMA does not directly measure viability or a quantitative relic-DNA fraction [36]. The experiment also did not provide a balanced comparison of the three soil positions, so it cannot show whether PMA-sensitive DNA was more abundant at the surface than below it.

The extraction blanks address a separate low-biomass risk: DNA introduced during laboratory processing. Candidate contamination was small in most linked samples, although one profile was strongly affected, and filtering left all 25 tracked conclusions unchanged (Figure 3d). This sensitivity supports the reported patterns for the linked profiles, but it cannot replace missing or assay-mismatched controls [15]. The control design and its trip-specific limits are reported in Materials and Methods and Supplementary Section S2.

A regional framework for the Arabian Peninsula

Earlier Arabian studies established local habitat-specific patterns. A groundwater-fed UAE sabkha contained a saline microbial mat, while Saudi studies examined highland soils, arid locations, biological crusts and desert plants [16, 17, 37, 13, 2]. This work revealed local habitat and plant effects without mapping the open interior. Our survey supplies that landscape frame and repeated soil-position measurements.

The mean pH of 7.986, measured in a calcium chloride solution, matches the alkaline range reported for Saudi arid soils [17]. At AlUla, a plant-created patch of more fertile soil had a different bacterial community from nearby bare sand; the bare sand was also more alkaline [14]. In the Empty Quarter, pH shared substantial variation with geography rather than replacing it. Plant patches may therefore modify a broader regional setting.

Regional experiments help interpret the soil-position and rainfall results. Around *Rhazya stricta*, soil compartment had a larger effect than simulated rain, while crust biomass increased 3-fold 1 week after watering [37]. At Jizan and Al Wahbah, rhizosphere and root communities differed from bulk soil, with the strongest loss of richness and evenness inside roots [13]. Our samples were loose soil near roots, not root-attached soil, so their lower evenness extends the observed local organization without identifying a host filter. The rapid watering result independently shows that an Arabian desert community can change within days, but its crust-biomass endpoint does not validate our bacterial-richness curve. Host identity, metabolite exchange and plant benefit remain open questions.

The mapped variation makes this survey useful for regional decisions because a single reference site or soil position could be mistaken for the state of the wider desert. Saudi Arabia's cloud-seeding programme aims to enhance rain and support ecosystem restoration [38]. Our rainfall association suggests a concrete sampling schedule for testing soil-community change during the first several days after natural or enhanced rain, while it does not predict the effect of cloud seeding. The western Rub' al-Khali also contains the Uruq Bani Ma'arid protected area and free-ranging Arabian oryx [39]; drought effects on reintroduced oryx show why habitat condition and climate matter to management [40]. Repeated microbial measurements could add soil and vegetation context before and after habitat restoration, but they are not a measure of oryx health. For arid agriculture, the survey identifies comparable background sites and candidate taxa or predicted pathways for later plant trials [17, 13]. The linked samples, measurements, sequences and provenance make these comparisons reusable [21]; they do not show that a microbial treatment will improve crops.

Limitations and identifying mechanisms

Our study primarily measures associations and does not establish causes. The sites lie on one west–east route, climate changes along that route, and every expedition travelled in the same direction. The shared route links location to collection order, time of day and changes during travel, and the study design cannot separate their individual effects from geography. Independent or crossing transects with randomized route direction would separate these alternatives, but they are difficult in the roadless dune interior. The field team must use long, logistically feasible off-road traverses through an isolated landscape, which limits route randomization and simultaneous crossing surveys [1]. A practical next design could reverse travel direction between expeditions, add shorter crossing transects where terrain allows, and leave autonomous soil sensors at selected sites.

Several measurements needed to test ecological explanations are missing. Air measurements are not soil microclimate, and XRF measures total elements rather than soluble ions, conductivity or osmotic stress [41]. The rainfall model used gridded daily estimates and collection dates, not soil moisture or collection time. Rain was too rare and uneven for omission of one expedition to be a meaningful stability test; the randomization test instead shifted the timing of the rain observed within each expedition. Its first-several-days window remains an association, and models that allowed more complex geographic change weakened the evidence. Event-triggered sampling before and after storms, moisture sensors and watering experiments are needed to measure response time and test a mechanism. Root identity, root distance, carbon, texture and plant performance would likewise separate plant effects from shared habitat.

Shotgun metagenomics could recover genes without biases caused by 16S primers or by bacteria carrying different numbers of 16S gene copies. It could also link functions to genomes, resolve strains, and include archaea, fungi and viruses. It would test whether predicted pathways occur in the local gene pool, but DNA still measures potential. Transcripts, proteins, metabolites, cell counts and process rates are needed to show activity, while manipulations remain necessary for causation [42].

Control coverage was uneven. Purified-DNA standards do not test extraction, whole-cell standards do, Trip 4 lacked a positive standard, and Trip 5 controls used shotgun rather than 16S. Only linked Trip 5 extraction blanks could train the contaminant sensitivity. Future work should sequence extraction and PCR blanks with every batch, use positive standards that enter at the same stage as the biological samples, and retain DNA concentrations. Stable filtered results cannot substitute for missing controls.

Conclusions

The Empty Quarter is a structured microbial landscape. Bacterial assemblages turn over across the sand sea, nearby soil positions differ in composition and evenness, environmental gradients track the broad pattern, and recent rain was associated with a short rise in richness. Predicted metabolic potential also changes with geography and soil position. These findings extend recurring dryland principles to the world's largest continuous sand desert while showing that the direction of diversity–environment associations remains local. The survey provides a regional reference and a set of testable hypotheses bounded by relic-DNA and contamination checks.

MATERIALS AND METHODS

Field design and samples used in each analysis

We sampled one transect through the Saudi Rub' al-Khali during expeditions on March 17–20, 2023 (T1), July 14, 2023 (T2), February 17–21, 2024 (T3), August 19–21, 2024 (T4), and October 12–15, 2025 (T5). Heat restricted T2 to 8 sites; the other campaigns covered the main route. At each site and visit, we sampled surface soil (0 cm), shallow subsurface soil (5–10 cm) and loose soil from the root zone at 5–10 cm, nominally in 3 co-located field replicates. Soil was placed in sterile tubes by scooping directly with the tube or with a sterile plastic spoon. Samples were kept at $-20\,^{\circ}$C in a portable freezer during each expedition and transferred to $-80\,^{\circ}$C in the laboratory. We use "root-adjacent soil" rather than rhizosphere because soil was not consistently attached to roots.

Here, a profile means one quality-controlled 16S amplicon library and its counts for exact bacterial 16S amplicon sequences, which we call amplicon sequence variants (ASVs). Analyses that required repeat visits focused on sites 1–60 and used 1,227 profiles. Four locations sampled only during Trip 1 contributed 10 more profiles. We retained them for description but excluded them from repeated-site analyses because later expeditions could not revisit them. Supplementary Table S1 reports the samples used in each analysis.

Amplicon processing

Between 0.9 and 1.2 g of soil was extracted with QIAGEN PowerLyzer PowerSoil or DNeasy PowerSoil Pro kits. Samples were homogenized with a PowerLyzer 24, processed manually or with a QIACube Connect, and eluted in 50 μL. The V3–V4 region of the 16S rRNA gene was amplified with the Bakt_341F (`CCTACGGGNGGCWGCAG`) and Bakt_785R (`GACTACHVGGGTATCTAATCC`) primers [43], both carrying Illumina overhang adapters. Amplification used the QIAGEN Multiplex PCR Kit: 95 °C for 15 min; 35 cycles of 94 °C for 30 s, 57 °C for 90 s and 72 °C for 90 s; and a final 72 °C extension for 10 min. Amplicons were checked on a 2 % agarose gel, indexed and cleaned following the Illumina 16S protocol [44], quantified with the Qubit high-sensitivity double-stranded DNA assay, sized on an Agilent TapeStation 4200, and pooled by concentration and size. Paired-end 2×250-base libraries were sequenced on an Illumina NovaSeq 6000. Reads were processed with nf-core/Ampliseq v2.14.0 [45, 46]. We used DADA2 to infer exact amplicon sequence variants (ASVs) and SILVA 138.2 [47] to assign taxonomy. The complete count table retained all 351,472 ASVs and 24 extraction blanks. We excluded these blanks and 10 biological profiles below 1,000 reads, leaving 1,237 profiles. Positive standards, PCR blanks and no-template controls were evaluated in their source-specific data and were never ecological profiles. The unfiltered biological table was the primary analysis input.

Assay controls and sensitivity analyses

We followed low-biomass guidance by analysing each control according to the stage it tested [15]. 17 sequenced extraction blanks could be linked by extraction day to 217 canonical Trip 5 profiles. We pooled these blanks and marked an ASV as a candidate when it occurred in at least 2 blanks and was more prevalent in blanks than in biological samples in a one-sided Fisher test ($p < 0.1$). The 351 candidates were removed only from their linked profiles in a filtered copy used to test sensitivity. 7 other extraction blanks lacked a day mapping and were characterized without being used to identify candidates. PCR blanks contain the amplification reagents but no sample, so they test a later laboratory step. PCR blanks were kept separate because their biological batch links were incomplete. Positive standards were used only to assess expected-taxon recovery and whether their expected taxa appeared in other samples. Supplementary Section S2 gives the trip-specific products, assay stages and remaining coverage gaps.

For the contamination sensitivity, we removed the 351 candidate ASVs only from the 217 linked Trip 5 profiles. Candidate sequences comprised 2.19 % of reads overall and a median of 0.40 % per profile (interquartile range, 0.16–0.99 %); 1 profile contained 56.60 % candidate reads (Figure 3d). Every filtered profile remained above 25,000 reads. We repeated 25 geographic, environmental and soil-position conclusions with this filtered copy; all retained the same disposition as in the unfiltered primary analysis.

Propidium monoazide (PMA) enters cells with damaged membranes and prevents their exposed DNA from being amplified [36]. We compared 9 matched treated and untreated Trip 5 aliquot pairs. We calculated Hurlbert expected richness at 123,897 reads, the minimum depth among the 18 libraries, without random subsampling, and calculated Shannon diversity at full depth. Exact two-sided paired Wilcoxon tests compared the two endpoints; 99,999 bootstrap samples of the matched aliquots gave uncertainty for their mean differences (seed 20260805). Expected richness was lower after treatment in 8 pairs (mean treated-minus-untreated difference, -163.3 taxa; 95 % paired-aliquot bootstrap interval, -274.0 to -69.8; exact paired $p = 0.00781$; Figure 3c). This experiment measures a PMA-sensitive richness difference in these aliquots. It does not estimate cell viability, a survey-wide relic-DNA fraction, or a difference among the 3 surveyed soil positions. Supplementary Sections S2 and S7.3 give the control provenance, quality checks and uncertainty calculations.

Environmental measurements

At each primary site visit we recorded air temperature and, where available, pressure and relative humidity. We retained confirmed values and excluded 1 out-of-range measurement. Supplementary models related this collection weather to diversity after accounting for campaign and for a geographic pattern that could change steadily or change its rate smoothly and reverse once along the route. Uncertainty estimates kept repeated measurements from the same site together. These are air, not soil, measurements.

We obtained site-specific monthly air temperature, total rain and relative humidity from the archived Open-Meteo historical API resource [24]. Each core site had 49 complete months from January 2022 to January 2026. We averaged climate and the 3 diversity measures within each site, then used the 60 sites for Spearman correlation. The 9 climate–diversity tests were corrected together with the Benjamini–Hochberg method, which limits the expected proportion of false discoveries among supported results. For abundance, we selected the 200 genera with the largest mean abundance among those detected in at least 20 % of profiles. We added 0.5 to every count so that zero counts could enter the logarithm, then calculated each genus's log ratio to the geometric mean abundance across all genera. This centred-log-ratio (CLR) transformation compares taxa within a relative composition instead of treating read counts as absolute abundances. We averaged the transformed abundances by site. The 600 climate–genus tests were corrected together in a second set.

We measured pH in 767 archived-soil specimens from all 5 expeditions. Dry soil below 2 mm was mixed with 0.01 M $CaCl_2$ at 1:2.5 mass-to-volume and measured with an accumet AB315. We required a valid date and range, recorded calibration, an electrode slope of 95–102 %, and a pH 10 buffer read-back of 9.9–10.1. In total, 712 measurements passed and 702 matched ecology profiles; we did not impute missing values. We averaged matched values within site, expedition and position. The composition test added pH after expedition, position and site and randomly rearranged the remaining variation within sites 999 times. We compared the transect result before and after pH adjustment in the same samples.

Laboratory XRF covered 725 archived-soil specimens. We retained 11 elements detected in at least 75 % of specimens, transformed values as $\log(1+x)$, placed them on a common scale within each expedition and combined them by principal component analysis. This method constructs weighted combinations of the elements; the first component captures the largest shared change. We tested whether this component explained bacterial composition beyond expedition, position and site, using 999 random rearrangements within sites. Changes to the included taxa and the treatment of zero values tested sensitivity. XRF describes total elements, not salinity or soluble ions [41]. Supplementary Sections S2–S4 give the laboratory quality checks, coverage and complete models.

Short-term rainfall response

We obtained daily precipitation for each site from the versioned NASA POWER `PRECTOTCORR` resource [25]; Open-Meteo provided an independent rainfall dataset for comparison. We excluded the sampling date because collection times were unavailable. Technical replicates were averaged within 617 site, expedition and soil-position groups.

We represented a temporary response by weighting rain over the 60 complete days before collection. For rain $r_{i,d}$ at site observation i and d complete days before collection, the pulse score for a possible peak at day p was

$$x_i(p) = \sum_{d=1}^{60} r_{i,d}(d/p)\exp(1-d/p).$$

The weight rises to one at $d = p$ and then declines. We searched $p = 0.5, 1.0, \ldots, 30.0$ days for expected richness at 25,000 reads, Shannon diversity and normalized evenness. Each model allowed each site and soil position to have its own baseline, each expedition to have its own mean, and longitude and collection day to have separate steady trends within each expedition.

To account for choosing the strongest result from many peaks and diversity measures, we shifted the full 60-day rain sequence in a circle 19,999 times, independently within each expedition (seed 20260804). Each shifted dataset retains the amount, rarity and spatial pattern of rain but changes its timing relative to collection. We used a one-sided test for the stated prediction of a positive pulse and also report a two-sided test that allows either direction; both tests cover the complete search. Confidence intervals for the selected effect used sandwich covariance clustered by site. We resampled whole sites 9,999 times (seed 20260805) to estimate the share of remaining variation explained by rain and the uncertainty in peak selection. Supplementary Section S7 describes these calculations and the comparison of rainfall datasets. Additional analyses changed the rainfall dataset, contaminant filtering, pH adjustment, geographic trend and included samples. To check the assumed pulse shape, we fitted rain from 1, 2, 3–4, 5–7, 8–14, 15–30 and 31–60 complete days before collection together for each rainfall dataset, with site-clustered uncertainty. We also used future rain as a negative control. Supplementary Section S7 gives these results and the exact reproducibility manifests.

Community, diversity and spatial analyses

To compare richness at the same sequencing depth, we calculated Hurlbert's expected richness: the number of taxa expected if each profile contributed 25,000 reads. Shannon diversity used full depth. We averaged profiles within site, expedition and position before paired position comparisons. Means and intervals came from repeatedly sampling whole sites with replacement. We drew 1,000,000 whole-site bootstrap samples (seed 20260723), keeping all repeated observations from one site together. We corrected each family of 3 contrasts. An additional evenness analysis used $H/\log(E[S_{25k}])$, where H is Shannon diversity and $E[S_{25k}]$ is expected richness at 25,000 reads. This measures abundance balance relative to common-depth richness; it is not conventional Pielou evenness. Supplementary Section S5.2 gives campaign-specific contrasts and correction families.

For composition, we combined replicate counts within expedition, site and position. We selected the 200 most abundant genera among those detected in at least 20 % of profiles, added 0.5 to every count to handle zeros, and used the CLR transformation defined above [22]. This small addition is called a pseudocount. The paired test used 170 complete three-position comparisons across 60 sites. Pairwise tests randomly reversed the direction of each within-pair difference while keeping all observations from a site together. Alternative taxon sets, treatments of zero counts and expedition omissions tested sensitivity.

For broad spatial structure, we retained core-site groups with at least 2,000 reads, removed the CLR mean for each expedition and position, and averaged within site. The route model allowed both steady change from west to east and a smooth change in rate that could reverse once, represented by straight-line and squared-location terms. It used the same 200 genera and 999 random rearrangements of whole sites. Taxon sets, trend shapes and expedition omissions tested sensitivity. To quantify sampling uncertainty in the explained share, we removed each of the 60 sites in turn, refitted the model, calculated the delete-one-site jackknife standard error, and formed a 95 % t interval with 59 degrees of freedom.

To measure distance decay within each soil position, we removed the campaign mean from the CLR composition at each of the 60 sites and related Aitchison dissimilarity to distance. We randomly reassigned location labels to whole sites 9,999 times across all soil positions, so the 1,770 pairs of sites were not treated as independent observations. For comparisons based only on whether a taxon was present, we pooled counts within site and position and randomly subsampled each pooled sample to the same 12,865 reads (seed 20260728). We partitioned Sørensen dissimilarity into taxon replacement and nestedness [23]. Delete-one-site jackknife intervals for the distance slopes and mean replacement shares removed every pair involving the omitted site.

Supplementary Section S5 gives the spatial prediction, ASV-resolution and dispersion diagnostics.

Predicted functional potential and shotgun comparison

PICRUSt2 v2.4.1 predicted KEGG Ortholog groups and MetaCyc pathways from the ASV tables [26]. We matched 1,227 core-site profiles, summed replicates within expedition, site and position, and ranked pathways detected in at least 20 % of groups. The primary analysis used the 200 with largest mean abundance, a 0.5 pseudocount and CLR coordinates. For geography, we removed expedition-by-position means and tested the route model that allowed the rate of change to vary smoothly and reverse once, using 9,999 random rearrangements of whole sites. The pathway geography interval used the same delete-one-site jackknife and 95 % t interval as the taxonomic route model. For soil position, we expressed each complete three-position comparison relative to its own mean and then averaged within site. The overall test randomly reassigned position labels within sites; pairwise tests randomly reversed the site-level differences. We corrected 3 profile comparisons together and 600 pathway comparisons together. Pathway sets, pseudocounts and expedition omissions tested sensitivity.

For each predicted profile, the weighted nearest-sequenced-taxon index (NSTI) summarized the distance between its ASVs and sequenced reference genomes. A smaller value means that the prediction requires less extrapolation. For an empirical comparison, we matched PICRUSt2 and shotgun-derived abundance for 3,471 KEGG Ortholog gene families in 125 samples, calculated a Spearman rank correlation within each sample, and also correlated the two community-average profiles. Supplementary Section S8 reports the uncertainty method and the limits of pathway and marker-gene predictions.

Cross-desert contextual analysis

We compared effects in 2 public Atacama datasets without merging their ASV counts. In QIITA study 10360 (PRJEB17617), technical replicates were averaged within 16 sites; Spearman correlations related diversity to soil humidity before and after adjustment for latitude, longitude and elevation. In PRJEB39249, 62 intracellular-DNA profiles were aggregated to 23 sampled depths. Three depth zones were compared in 200-genus CLR coordinates with 9,999 label rearrangements; diversity was averaged after subsampling each profile 100 times to 8,000 reads. Supplementary Section S10 gives the upstream processing and comparison limits.

Sensitivity analyses and reproducibility

To test whether sequencing depth or recorded extraction method explained the Shannon position differences, we fitted a Gaussian generalized estimating equation with sequencing depth, expedition and extraction protocol. This regression method accounts for repeated observations from the same site. We treated these models as robustness checks because extraction protocol and laboratory batch were incompletely recorded and partly linked to expedition. The companion data descriptor covers the field, specimen, amplicon, environmental, XRF and knowledge-graph resources [21]. Inputs, scripts, machine-readable results, random seeds and checksums are maintained in the dedicated reproducibility repository at `https://github.com/bio-ontology-research-group/empty-quarter-ecology-reproducibility`. The repository links each analysis to its exact input, machine-readable result, random seed and checksum, and links shared data products to the companion resource. Public deposition of the ecology package and upstream amplicon, PMA, shotgun-read and genome resources remains a submission requirement.

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